Online Retail Sales Analysis Using SQL (PostgreSQL)

Project Overview

This project focuses on analyzing an online retail sales dataset using **PostgreSQL SQL queries** to extract meaningful business insights.

The objective of this project is to understand sales performance, customer behavior, product performance, country-wise contribution, and monthly sales trends using SQL-based data analysis.

The project demonstrates the ability to perform:

- Data validation and quality checks
- Exploratory Data Analysis (EDA)
- Business KPI calculations
- Customer segmentation analysis
- Product performance analysis
- Geographic sales analysis
- Time-series sales analysis

Dataset Description

The dataset contains online retail transaction records with details about:

Column	Description
invoice_no	Unique transaction/order number
stock_code	Product identification code
description	Product name
quantity	Number of units sold
invoice_date	Date and time of purchase
unit_price	Price per product unit
customer_id	Unique customer identifier
country	Customer location
sales	Total sales value (Quantity × Unit Price)

Column	Description
year	Transaction year
month	Transaction month
month_number	Month sorting number
quarter	Sales quarter
day	Day of transaction
hour	Purchase hour
customer_status	Customer category
market_type	UK or International market

Tools & Technologies Used

- **Database:** PostgreSQL
- **Language:** SQL
- **Data Analysis Techniques:**
- Aggregations
- GROUP BY
- ORDER BY
- CASE statements
- Subqueries
- Window functions
- Date-based analysis
- Ranking functions

Project Workflow

1. Data Validation & Quality Checks

Performed multiple checks to ensure data accuracy:

- Checked total number of records
- Verified column structure and data types
- Checked missing values
- Validated date range
- Identified negative quantity records
- Checked invalid sales values

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2. Key Business KPIs Analyzed

The project calculated important business metrics:

- Total Revenue
- Total Orders
- Total Quantity Sold
- Total Customers
- Average Order Value (AOV)
- Average Revenue per Customer
- Average Quantity Purchased per Order

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3. Customer Analysis

Analyzed customer purchasing behavior:

- Identified top customers by revenue
- Calculated repeat customers
- Analyzed customer distribution by country
- Measured customer contribution to revenue

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4. Product Analysis

Analyzed product performance:

- Top 10 products by revenue
- Top 10 products by quantity sold
- Identified best-performing products based on customer demand

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5. Country Analysis

Performed geographic sales analysis:

- Top revenue-generating countries
- Countries with highest order volume
- Country-wise average order value
- Revenue contribution percentage by country

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6. Time-Based Sales Analysis

Analyzed sales patterns over time:

- Monthly revenue trends
- Highest-performing months
- Year-wise sales performance
- Monthly revenue growth comparison
- Average monthly order value

Key Business Insights

1. Overall Business Performance

- The business generated approximately **£10.64 million in total revenue**.
- More than **524K transactions/orders** were analyzed.
- The dataset contains millions of product units sold, showing strong customer demand.
- Average Order Value analysis helps understand customer spending behavior.

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2. Customer Insights

- A small group of customers contributed a significant portion of total revenue.
- Repeat customers represent valuable buyers who generate consistent sales.
- Customer analysis can help businesses create loyalty programs and personalized marketing campaigns.

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3. Product Insights

- A few top-selling products generated a large percentage of revenue.
- Products with high quantity sales indicate strong customer demand.
- Businesses can focus inventory planning on high-performing products to improve profitability.

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4. Geographic Insights

- The United Kingdom was the strongest market in terms of revenue and order volume.
- International markets contributed additional revenue opportunities.
- Country-level analysis helps identify expansion opportunities.

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5. Time-Based Insights

- Sales performance changed throughout the year, showing seasonal buying patterns.
- Monthly revenue analysis helps identify peak sales periods.
- Revenue growth comparison helps evaluate business improvement over time.

Business Recommendations

Based on SQL analysis:

1. Focus marketing campaigns on high-value and repeat customers.
2. Maintain sufficient inventory for top-performing products.
3. Analyze seasonal trends to optimize sales campaigns.
4. Explore international markets with increasing revenue contribution.
5. Use customer purchase behavior to create targeted offers.

SQL Analysis Questions Covered

This project answers 29 business questions including:

KPI Analysis

- What is total revenue?
- How many orders were placed?
- How many customers purchased?
- What is the average order value?

Customer Analysis

- Who are the top customers?
- Which customers are repeat buyers?
- How many customers exist in each country?

Product Analysis

- What are the top products by revenue?
- What products sell the highest quantity?

Country Analysis

- Which countries generate the highest revenue?
- Which countries have the most orders?
- What percentage of revenue comes from each country?

Time Analysis

- Which month performed best?
- What are monthly sales trends?
- How does revenue change month over month?

Project Outcome

This project demonstrates practical SQL skills required for a Data Analyst role, including:

✅ Data cleaning validation
✅ Business metric calculation
✅ Advanced SQL querying
✅ Customer and product analysis
✅ Sales trend analysis
✅ Translating raw data into business insights

Author

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Aspiring Data Analyst skilled in:

- Excel
- SQL
- Power BI
- Data Visualization
- Business Analytics